

Work
Energy
Power



Applied Physics

Work





★ When you apply a force on any body it covers displacement then work is said to be done on the body.

★ Pulling a Wagon

★ Climbing stairs

★ Falling Down

★ Carrying a Heavy Backpack Down the Hall



✧ Which of the following is NOT work?

✧ Pushing a Stalled Car

✧ Pulling a Wagon

✧ Climbing stairs

✧ Falling Down

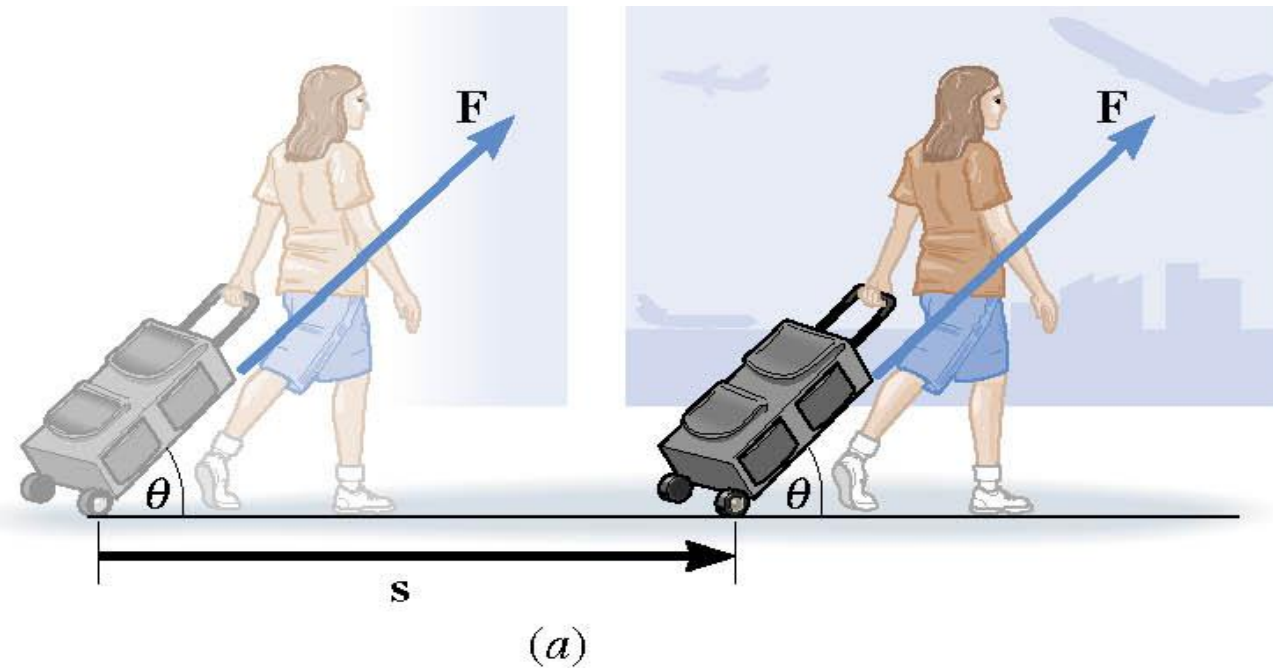
✧ Carrying a Heavy Backpack Down the Hall



6-01 Work and Power

✳ Sobia pulls a backpack on wheels down the 100-m hall. The 60-N force is applied at an angle of 30° above the horizontal. How much work is done by Sobia?

✳ $W = 5196 \text{ J}$



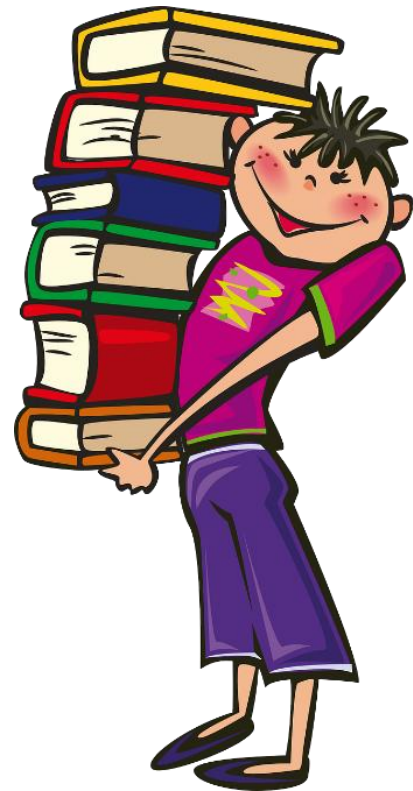


6-01 Work and Power

✧ Drew is carrying books (200 N) down the 100-m hall. How much work is Drew doing on the books?

✧ $W = 0 \text{ J}$

✧ The force is vertical
displacement is horizontal.





6-01 Work and Power

✧ You carry some books (200 N) while walking down stairs height 2 m and length 3 m. How much work do you do?

$$\star W = -400 \text{ J}$$





6-01 Work and Power

- ✧ A suitcase is hanging straight down from your hand as you ride an escalator. Your hand exerts a force on the suitcase, and this force does work. Which one of the following is correct?
- ✧ The W is negative when you ride up and positive when you ride down
- ✧ The W is positive when you ride up and negative when you ride down
- ✧ The W is positive
- ✧ The W is negative



✧ Two cars with the same mass do the same amount of work to get to 100 km/h.

✧ Which car is better

✧ Takes 8.0 s

✧ Takes 6.2 s

✧ Sometimes the time taken to do the work is important



✧ Rate that work is done

$$P = \frac{W}{t}$$

✧ Unit: joule/s = **watt (W)**



6-01 Work and Power

- ✧ Since work changes the amount of energy in an object
- ✧ Power is the rate that energy is changing



6-01 Work and Power

✧ A 1000 kg car accelerates from 0 to 100 km/h in 3.2 s on a level road. Find the average power of the car.

✧ $P = 119889.06 \text{ W}$





✧ Electrical Energy

✧ Often measured in kWh because $Pt = W$

✧ If it costs \$0.10 per kWh, how much will it cost to run a 1000 W microwave for 2 minutes?



- ✧ Calculate the electricity bill amount for a month of 31 days, if the following devices are used as specified:
- a) 3 bulbs of 30 watts for 5 hours
 - b) 4 tube lights of 50 watts for 8 hours
 - c) 1 fridge of 300 watts for 24 hours
- Given the rate of electricity is 2 Rs. per unit



✧ Solution

The energy consumed by the bulbs,

As we know energy=power×time

$$\mathbf{3 \text{ bulbs} \times 30 \text{ watts} \times 5 \text{ hours} \times 31 \text{ days} = 13950 \text{ Wh}}$$

The energy consumed by the tubes,

$$\mathbf{4 \text{ tubes} \times 50 \text{ watts} \times 8 \text{ hours} \times 31 \text{ days} = 49600 \text{ Wh}}$$

The energy consumed by the fridge,

$$\mathbf{1 \text{ fridge} \times 300 \text{ watts} \times 24 \text{ hours} \times 31 \text{ days} = 223200 \text{ Wh}}$$

Therefore, the total energy consumption is given by,

$$\mathbf{13950+49600+223200 = 286750 \text{ Wh} = 286.75 \text{ KWh}}$$

We need to convert it into units, where 1 unit = 1 kWh

$$\mathbf{\text{So, electricity bill} = 286.75 \text{ units} \times 2 \text{ rs} = \text{Rs. } 573.5}$$

In this lesson you will...

- Find kinetic energy.
- Find potential energy.

Types of Energy





✧ Energy is the ability to do work

✧ Kinetic Energy - Energy due to motion

✧ If something in motion hits an object, it will move it some distance



6-02 Types of Energy

- ✧ Potential energy

- ✧ Energy due to position

- ✧ $W = Fd$

- ✧ Gravity

- ✧ $W_{gravity} = mgh$

$$PE = mgh$$

- ✧ Since the force of gravity is down

- ✧ We only worry about the vertical distance

- ✧ Potential Energy is not absolute

- ✧ It is a difference

- ✧ The path the object takes doesn't matter, just the vertical distance

- ✧ h is measured from any chosen point. Just be consistent



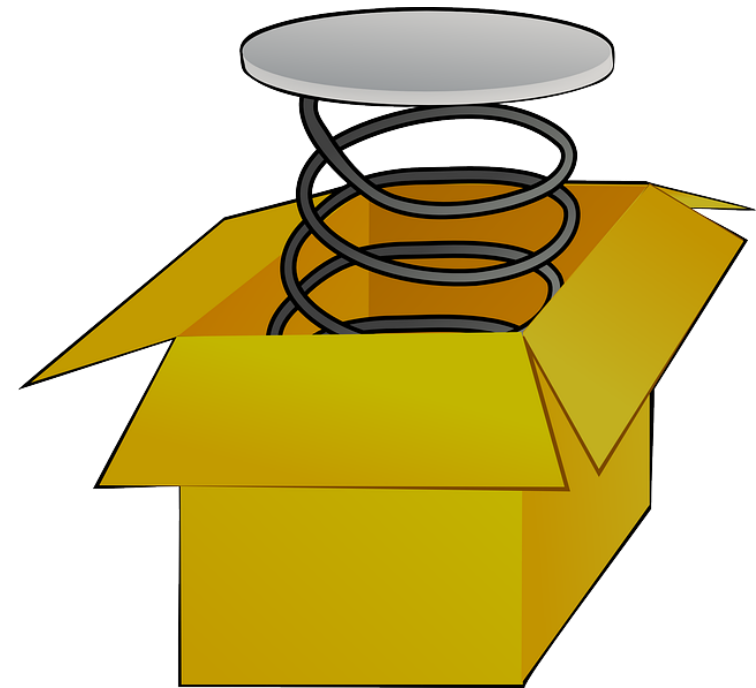
6-02 Types of Energy

★ Spring Potential Energy

$$★ W = Fd$$

★ $F = kx$ and $d = x$, but it requires calculus to properly calculate the work because the size of the force changes with the distance.

$$PE_s = \frac{1}{2} kx^2$$



6-02 Types of Energy

✧ A 5.2-kg Canada goose is flying towards you at 18 m/s and a height of 3 m. What is its (a) kinetic energy and (b) potential energy?





6-02 Types of Energy

✧ Let's say a coil suspension spring on a car is compressed 9.0 cm to after it is installed in a car. If it has a spring constant of 33000 N/m, what is the potential energy stored in the spring?



In this lesson you will...

- Convert energy from one form to another

Mechanical Energy Conservation



Conservation of Mechanical Energy of a Free falling object

$$\text{Total Mechanical Energy} = K.E. + P.E.$$

Work done to lift object to height $h = P.E$

At point A velocity = 0 $\therefore K.E.(A) = \frac{1}{2}mv^2 = 0$

Work done = $f \times s = (m \times g) \times h = mgh = P.E.$

$\therefore \text{Total M.E at A} = K.E. + P.E. = 0 + mgh = mgh$

height = h

$d(AB) = x$

Let velocity of object at point B = v_B

$$v_B^2 = u^2 + 2as = 0 + 2gx = 2gx$$

$$\therefore K.E.(B) = \frac{1}{2}mv^2 = \frac{1}{2} \times m \times 2gx = mgx$$

$$P.E. = mg(h - x) = mgh - mgx$$

$\therefore \text{Total M.E at B} = K.E. + P.E. = mgx + mgh - mgx = mgh$

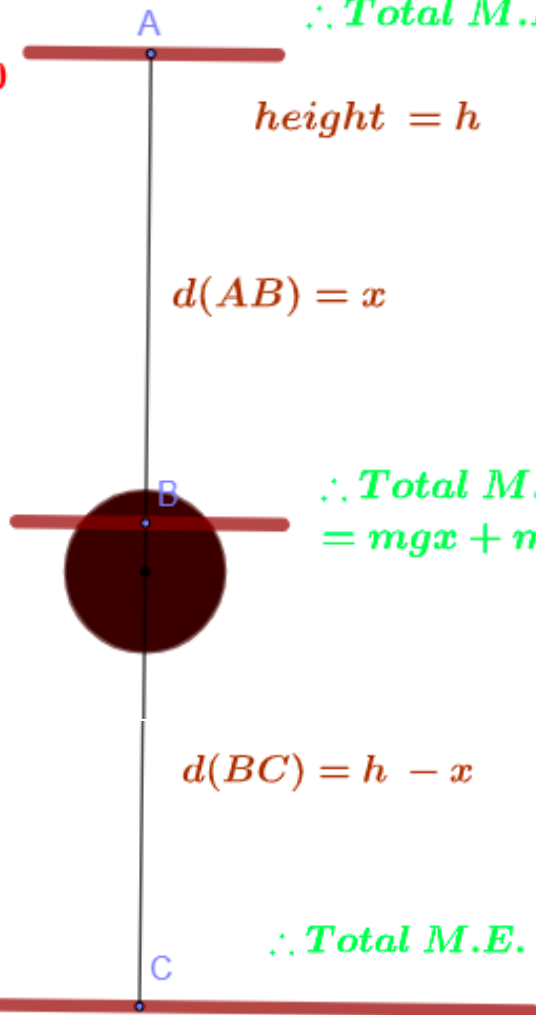
$$\text{velocity at point C} = u^2 + 2as = 0 + 2gh = 2gh$$

$$\therefore K.E(C) = \frac{1}{2}mv^2 = \frac{1}{2} \times m \times 2gh = mgh$$

At point C height = 0

$$\textcircled{P} P.E. = m \times g \times h = m \times g \times 0 = 0$$

$\therefore \text{Total M.E. at C} = K.E. + P.E = mgh + 0 = mgh$





6-03 Mechanical Energy Conservation

✧ Potential energy can be converted into Kinetic energy and back

✧ Think of an object thrown up

✧ Bottom $\rightarrow$ 0 PE, high KE

✧ Top $\rightarrow$ high PE, 0 KE

$$\star \Delta KE = -\Delta PE$$

$$\star KE_f - KE_0 = -(PE_f - PE_0)$$

✧ Rearrange

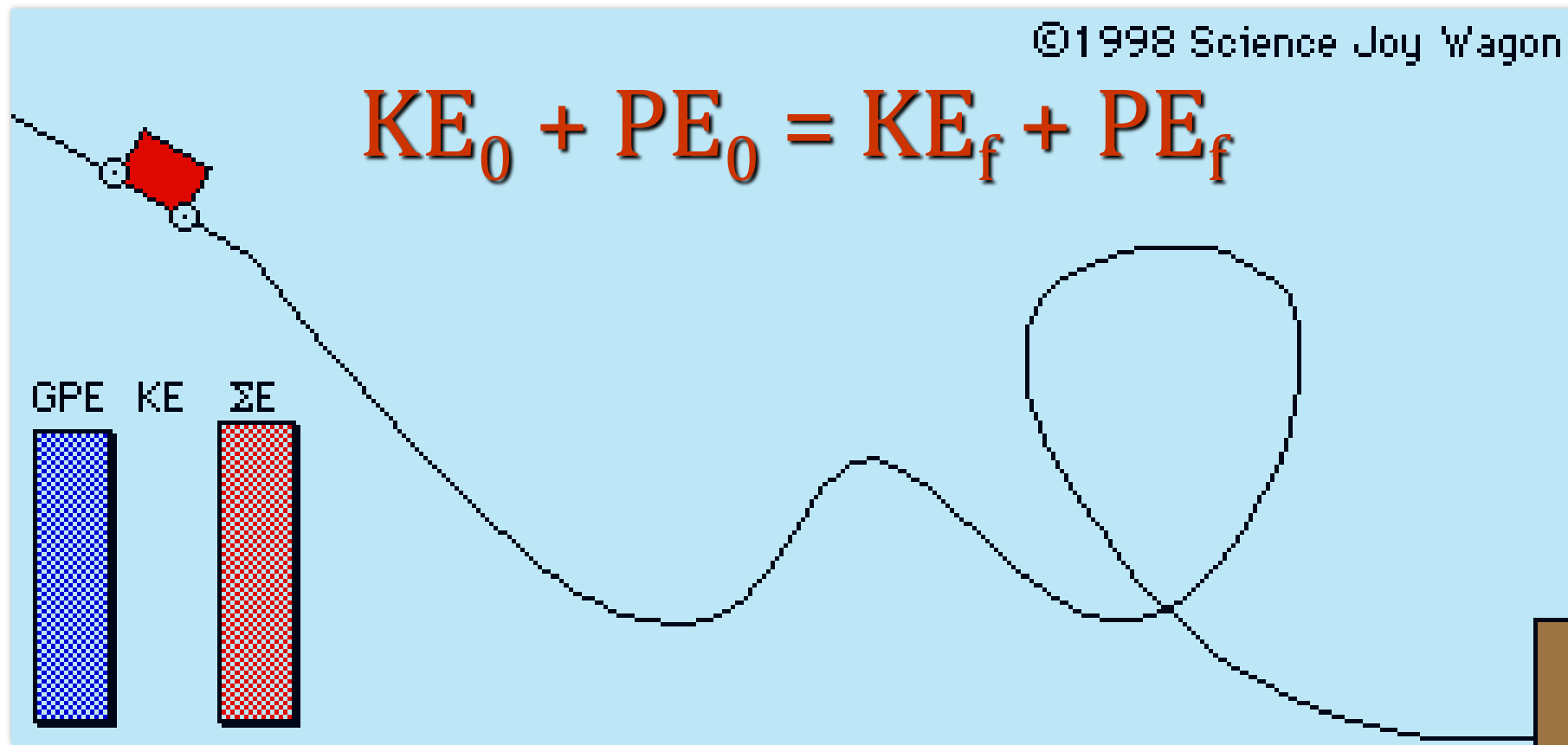
✧ Conservation of Mechanical Energy

$$KE_f + PE_f = KE_0 + PE_0$$



6-03 Mechanical Energy Conservation

- ✧ If there is only kinetic and potential energy
- ✧ Total mechanical energy is constant





- ★ A toy gun uses a spring to shoot plastic balls ($m = 50 \text{ g}$). The spring is compressed by 3.0 cm. Let $k = 2.22 \times 10^5 \text{ N/m}$.
- ★ (a) Of course, you have to do some work on the gun to arm it. How much work do you have to do?
- ★ (b) Suppose you fire the gun horizontally. How fast does the ball leave the gun?
- ★ (c) Now suppose you fire the gun straight upward. How high does the ball go?



✧ A 1500-kg car is driven off a 50-m cliff during a movie stunt. If it was going 20 m/s as it went off the cliff, how fast is it going as it hits the ground?

In this lesson you will...

- Convert energy from one form to another with work

Work and Conservation of Energy





✧ We can write Work done by net external force as

$$\star W_{net} = \Delta KE + \Delta PE$$

$$\star KE_0 + PE_0 + W_{net} = KE_f + PE_f$$

$$\star E_0 + W_{net} = E_f$$



6-04 Work and Conservation of Energy

★ Law of Conservation of Energy

- ★ The total energy is constant in any process. It may change form or be transferred from one system to another, but the total remains the same

★ Energy is transformed from one form to another

- ★ Box sliding down incline
 - ✦ PE transformed to KE
 - ✦ KE transformed to Heat and Sound
- ★ Engine
 - ✦ Chemical to KE and Heat



6-04 Work and Conservation of Energy

✧ A rocket starts on the ground at rest. Its final speed is 500 m/s and height is 5000 m. If the mass of the rocket stays approximately 200 kg. Find the work done by the rocket engine.

✧ $W = 3.48 \times 10^7 \text{ J}$





6-04 Work and Conservation of Energy

✧ A 1500-kg car's brakes failed and it coasts down a hill from rest. The hill is 10 m high and the car has a speed of 12 m/s at the bottom of the hill. How much work did friction do on the car?

$$\star W_f = -39000 \text{ J}$$





6-04 Work and Conservation of Energy

✧ Captain Proton's rocket pack provides 800,000 J of work to propel him from resting on his ship which is near the earth to 50 m above it. Captain Proton's mass is 90 kg. What is his final velocity?

✧ $v = 130 \text{ m/s}$

